EXPERIMENT-13:

Implement the Car Price Prediction Model using Python.

Code:

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# ============================================================
# Car Price Prediction using Linear Regression
# ============================================================
# Step 1: Upload Dataset
from google.colab import files
uploaded = files.upload()
# Step 2: Import Libraries
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import LabelEncoder
from sklearn.linear_model import LinearRegression
from sklearn.metrics import mean_absolute_error, mean_squared_error, r2_score
# Step 3: Load Dataset
df = pd.read_csv("CarPrice_Assignment.csv")
print("First 5 Rows")
print(df.head())
print("\nDataset Shape:", df.shape)
# Step 4: Remove Car_ID
if "car_ID" in df.columns:
    df.drop("car_ID", axis=1, inplace=True)
# Step 5: Convert Categorical Columns into Numeric
le = LabelEncoder()
for col in df.columns:
    if df[col].dtype == "object":
        df[col] = le.fit_transform(df[col])
```

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# Step 6: Features and Target
X = df.drop("price", axis=1)
y = df["price"]
# Step 7: Split Dataset
X_train, X_test, y_train, y_test = train_test_split(
    X,
    y,
    test_size=0.2,
    random_state=42
)
# Step 8: Train Model
model = LinearRegression()
model.fit(X_train, y_train)
# Step 9: Prediction
y_pred = model.predict(X_test)
# Step 10: Evaluation
print("\nModel Performance")
print("----------------------------")
print("R2 Score :", round(r2_score(y_test, y_pred),4))
print("MAE      :", round(mean_absolute_error(y_test, y_pred),2))
print("MSE      :", round(mean_squared_error(y_test, y_pred),2))
print("RMSE     :", round(np.sqrt(mean_squared_error(y_test, y_pred)),2))
# Step 11: Actual vs Predicted Plot
plt.figure(figsize=(8,6))
plt.scatter(y_test, y_pred)
plt.xlabel("Actual Price")
plt.ylabel("Predicted Price")
plt.title("Actual vs Predicted Car Prices")
plt.grid(True)
plt.show()
# Step 12: Predict Price of a New Car
sample = X.iloc[[0]]
```

prediction = model.predict(sample)

print("\nPredicted Price for Sample Car: ₹", round(prediction[0],2))

Output:

```
Choose Files  CarPrice_Assignment.csv
CarPrice_Assignment.csv(text/csv) - 26717 bytes, last modified: 8/5/2026 - 100% done
Saving CarPrice_Assignment.csv to CarPrice_Assignment (1).csv
First 5 Rows
   car_ID  symboling                   CarName fueltype aspiration doornumber  \
0       1          3        alfa-romero giulia      gas        std        two
1       2          3       alfa-romero stelvio      gas        std        two
2       3          1  alfa-romero Quadrifoglio      gas        std        two
3       4          2               audi 100 ls      gas        std       four
4       5          2                audi 100ls      gas        std       four

       carbody drivewheel enginelocation  wheelbase  ...  enginesize  \
0  convertible        rwd          front       88.6  ...         130
1  convertible        rwd          front       88.6  ...         130
2    hatchback        rwd          front       94.5  ...         152
3        sedan        fwd          front       99.8  ...         109
4        sedan        4wd          front       99.4  ...         136

   fuelsystem  boreratio  stroke compressionratio horsepower  peakrpm citympg  \
0        mpfi       3.47    2.68              9.0        111     5000      21
1        mpfi       3.47    2.68              9.0        111     5000      21
2        mpfi       2.68    3.47              9.0        154     5000      19
3        mpfi       3.19    3.40             10.0        102     5500      24
4        mpfi       3.19    3.40              8.0        115     5500      18

   highwaympg    price
0          27  13495.0
1          27  16500.0
2          26  16500.0
3          30  13950.0
4          22  17450.0

[5 rows x 26 columns]

Dataset Shape: (205, 26)

Model Performance
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R2 Score : 0.8407
MAE      : 2136.78
MSE      : 12575220.83
RMSE     : 3546.16
```

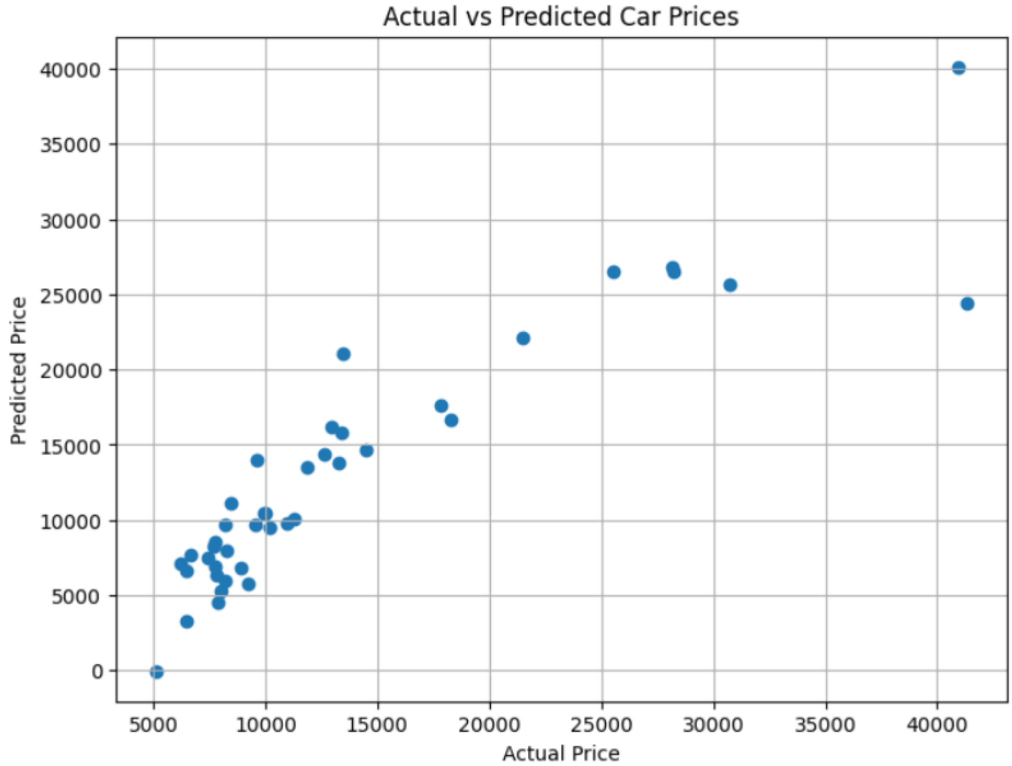


```
Predicted Price for Sample Car: ₹ 15308.46
```